Ravi Sharma

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Summary

Full-Stack & Applied AI Engineer specializing in scalable backend architecture and production-grade RAG systems. Proven experience in optimizing system latency, deploying vector databases, and engineering reliable, high-performance software.

Experience

Software Engineer Intern October 2026 – Present
7thGear

- Selected as Software Engineer Intern after being the only one among 10+ hackathon participants to deliver a working Google Meet bot that captures per-participant audio and video.
- Scoping the production Bot Streaming System for 20+ participant meetings, defining 5 outputs for the downstream analytics platform: speaker-isolated audio, per-person video, diarized transcript, join/leave metadata and live or post-call delivery.
- Designed a 4-pipeline identity architecture (roster, video binding, audio binding, full-roster exposure) and 3 decoupled stages (live capture, FFmpeg stitching, Cloudinary publishing) so post-processing never slows real-time recording.

Full-Stack Engineer March 2025 – May 2026
Strengthverse Private Limited

- Delivered 3 production-ready full-stack features (User Settings, Feedback, Sports Articles) on a React Native + Node.js/Express monolithic MVC application, engineered to support 350+ concurrent users on these features.
- Built and maintained Node.js/Express REST APIs on MongoDB for these features, monitoring production performance with Grafana and resolving live issues to keep them stable and responsive under load.
- Contributed to scaling the overall platform's supported user capacity from 50,000 to 800,000+ and concurrent capacity from 100 to 1,000+ users through high-quality feature engineering and production-focused development.

Projects

Nexdoc - Offline AI Doc Search | *Python, FastAPI, LanceDB, Electron* GitHub | Demo Video | Nexdoc.exe

- Architected a producer-consumer ingestion pipeline utilizing Watchdog and an auto-scaling, CPU-aware worker pool, enabling non-blocking concurrent processing across 5 document formats.
- Engineered modular extraction workflows integrating text normalization and sentence-aware chunking, achieving 88% extraction accuracy while persisting embeddings and enriched metadata into LanceDB and SQLite.
- Accelerated query latency from 5 seconds to under 1 second via LanceDB vector search, while maximizing retrieval accuracy through a multi-stage hybrid filtering, re-ranking, and dynamic context-windowing pipeline.
- Deployed a local-first desktop application featuring an optional Gemini-powered AI mode with strict citation grounding, anchoring 100% of synthesized answers directly to the top-5 verifiable document chunks.

MeetIngest - Attributed Meeting Data Bot | *Node.js, Playwright, WebRTC, FFmpeg* GitHub | Demo Video

- Developed Playwright bot hooking Meet WebRTC to record each participant as synced .mp4 (4-user tested)
- Mapped anonymous streams to names via DOM SSRC for video and speaker-to-audio correlation within 120 ms
- Kept recordings gap-free across re-negotiations (˜40% of meetings) via dual-recorder hand-off and FFmpeg
- Cut shutdown from ˜8 min to 5–25 s with a 3-phase traffic-light brake that finalizes all files first

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL
Frontend: React Native, React.js, Next.js, Tailwind CSS, Redux Toolkit, Axios, Zod, HTML5, CSS
Backend: Spring Boot, Node.js, Express.js, FastAPI, REST APIs, Spring Security, Hibernate, JPA, WebSockets
Databases: MongoDB, MySQL, PostgreSQL, LanceDB, SQLite
Tools & Platforms: Git/GitHub, Docker, Postman, Electron, Grafana, Playwright, FFmpeg, Cloudinary, AWS EC2
Concepts: DSA, RAG, Event-Driven Architecture, WebRTC, Real-Time Media Streaming

Education

Techno India University Kolkata, India
Master of Computer Applications (MCA) *August 2024 – June 2026*
Maulana Abul Kalam Azad University of Technology (MAKAUT) Kolkata, India
Bachelor of Computer Applications (BCA) *August 2021 – May 2024*